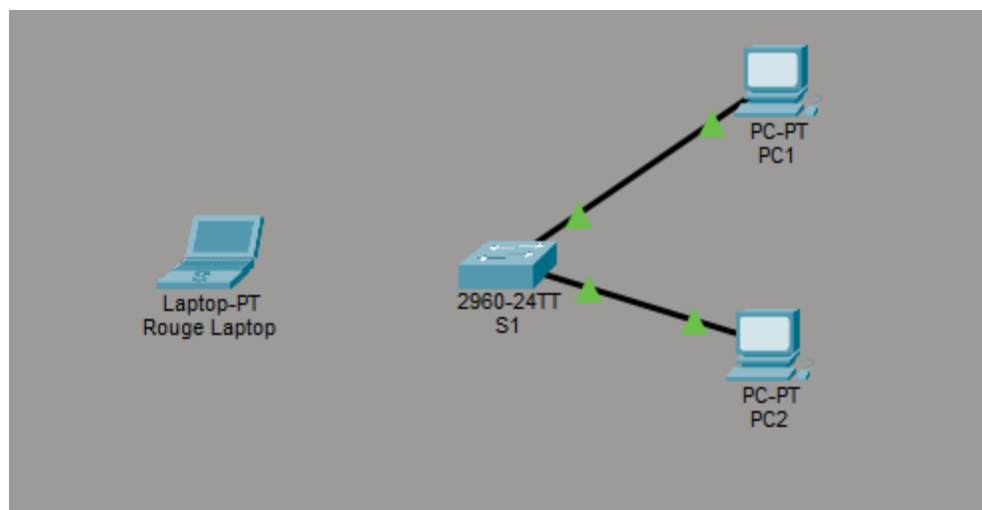


NAME: Carbonel, Nicole D.G.	Date: June 20, 2026
SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Implement Port Security

Addressing Table

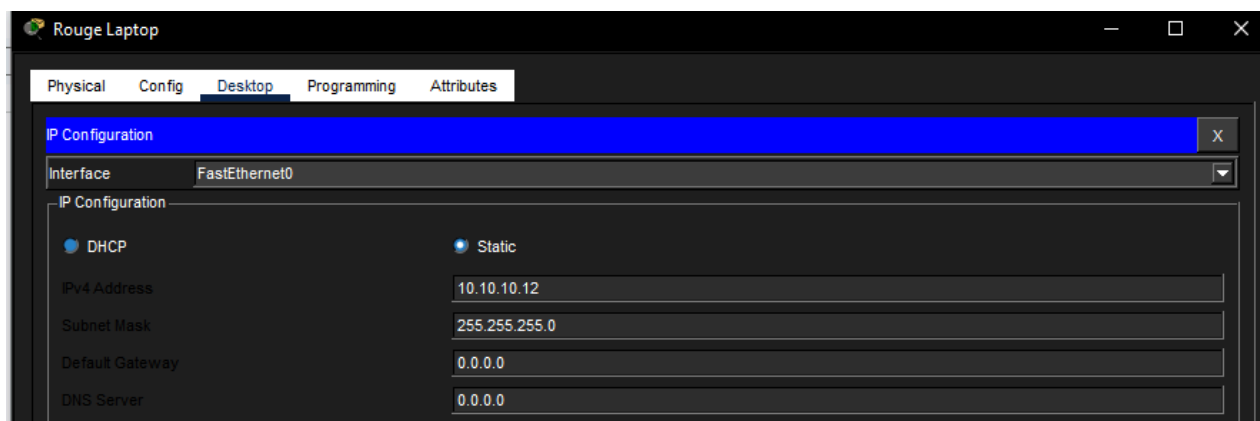
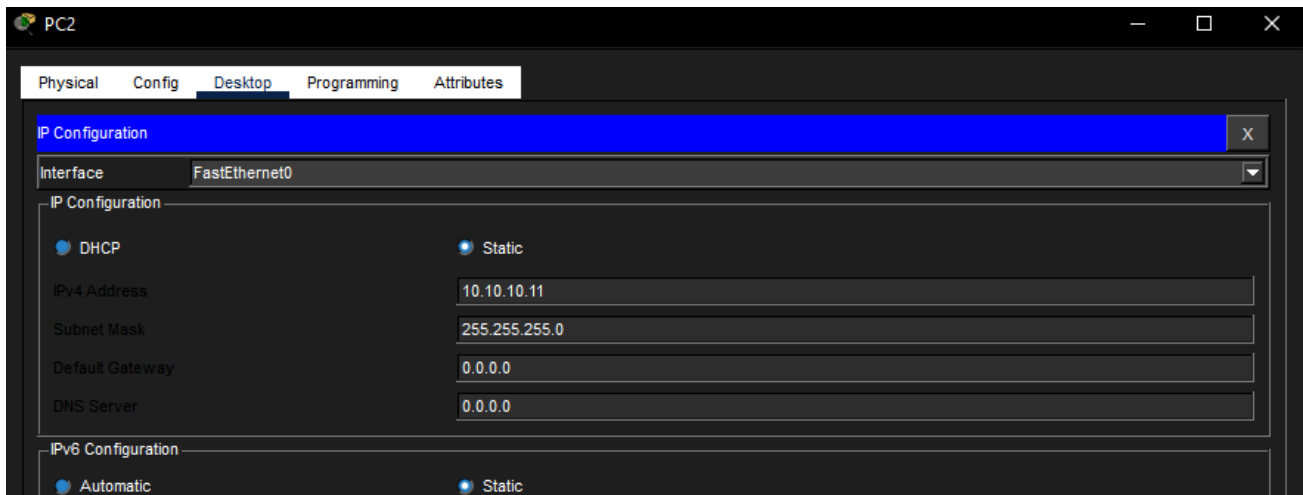
Device	Interface	IP Address	Subnet Mask
S1	VLAN 1	10.10.10.2	255.255.255.0
PC1	NIC	10.10.10.10	255.255.255.0
PC2	NIC	10.10.10.11	255.255.255.0
Rogue Laptop	NIC	10.10.10.12	255.255.255.0



Packet Tracer - Implement Port Security

```
S1>ena
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int vlan 1
S1(config-if)#ip add 10.10.10.2 255.255.255.0
S1(config-if)#no shut

S1(config-if)#
```



Objective

Part 1: Configure Port Security

Part 2: Verify Port Security

Background

In this activity, you will configure and verify port security on a switch. Port security allows you to restrict a port's ingress traffic by limiting the MAC addresses that are allowed to send traffic into the port.

Part 1: Configure Port Security

- a. Access the command line for **S1** and enable port security on Fast Ethernet ports 0/1 and 0/2.

```
S1(config)# interface range f0/1 - 2
```

```
S1(config-if-range)# switchport port-security
```

- b. Set the maximum so that only one device can access the Fast Ethernet ports 0/1 and 0/2.

```
S1(config-if-range)# switchport port-security maximum 1
```

- c. Secure the ports so that the MAC address of a device is dynamically learned and added to the running configuration.

```
S1(config-if-range)# switchport port-security mac-address sticky
```

Note: Set the violation mode so that the Fast Ethernet ports 0/1 and 0/2 are not disabled when a violation occurs, but a notification of the security violation is generated and packets from the unknown source are dropped.

```
S1(config-if-range)# switchport port-security violation restrict
```

Note: Disable all the remaining unused ports. Use the **range** keyword to apply this configuration to all the ports simultaneously.

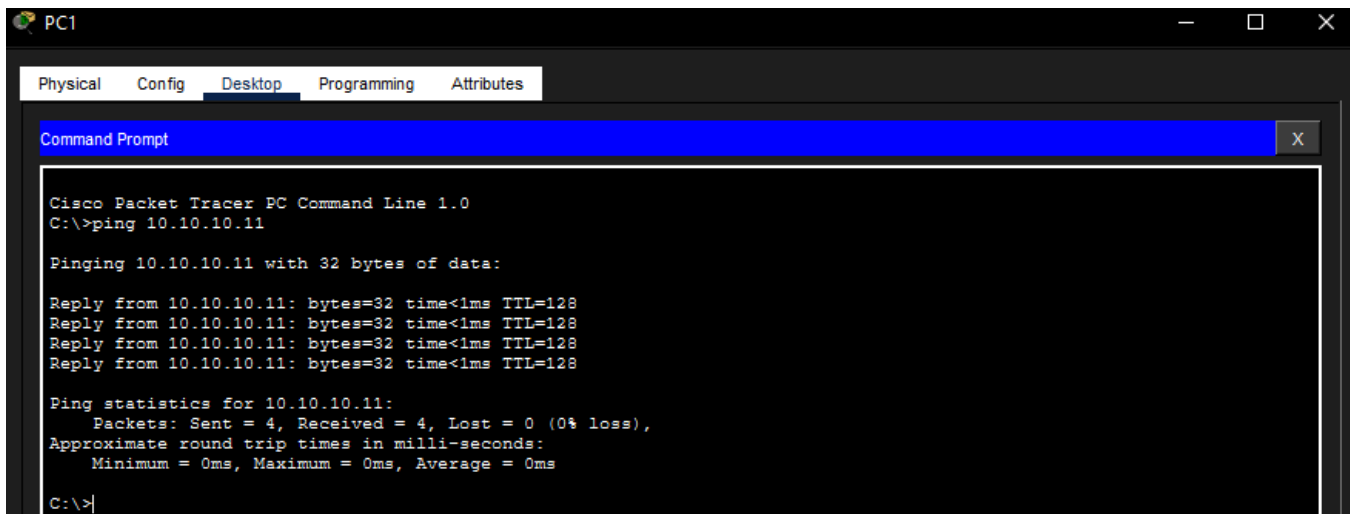
```
S1(config-if-range)# interface range f0/3 - 24, g0/1 - 2
```

```
S1(config-if-range)# shutdown
```

Note: Verify Port Security

```
S1>ena
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int range f0/1 - 2
S1(config-if-range)#swit
S1(config-if-range)#switchport mode access
S1(config-if-range)#swit
S1(config-if-range)#switchport port-security
S1(config-if-range)#swi
S1(config-if-range)#switchport port-sec
S1(config-if-range)#switchport port-security maximum 1
S1(config-if-range)#switc
S1(config-if-range)#switchport port-secu
S1(config-if-range)#switchport port-security mac-address sticky
S1(config-if-range)#swit
S1(config-if-range)#switchport port-se
S1(config-if-range)#switchport port-security violation restrict
S1(config-if-range)#int range f0/3 - 24, g0/1 - 2
S1(config-if-range)#shutdown
```

- d. From **PC1**, ping **PC2**.



- e. Verify that port security is enabled and the MAC addresses of **PC1** and **PC2** were added to the running configuration.

```
S1# show run | begin interface
```

Note: Use port-security show commands to display configuration information.

```
S1#show run | begin interface
interface FastEthernet0/1
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security violation restrict
  switchport port-security mac-address sticky 00D0.FF81.10A0
!
interface FastEthernet0/2
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security violation restrict
  switchport port-security mac-address sticky 0000.0C53.0680
!
interface FastEthernet0/3
  shutdown
!
interface FastEthernet0/4
  shutdown
.
```

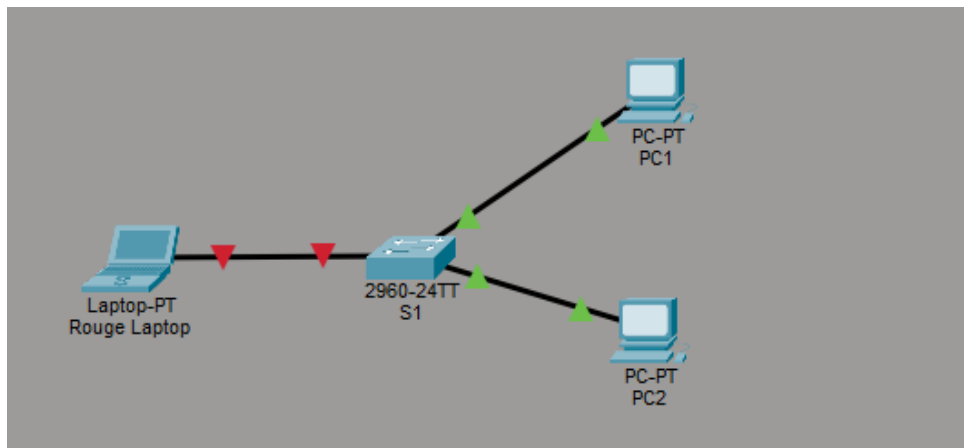
S1# **show port-security**

```
S1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/1      1              1              0          Restrict
Fa0/2      1              1              0          Restrict
-----
```

S1# **show port-security address**

```
S1#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
      (mins)
-----
1       00D0.FF81.10A0   SecureSticky        Fa0/1    -
1       0000.0C53.0680   SecureSticky        Fa0/2    -
-----
```

Note: Attach **Rogue Laptop** to any unused switch port and notice that the link lights are red.

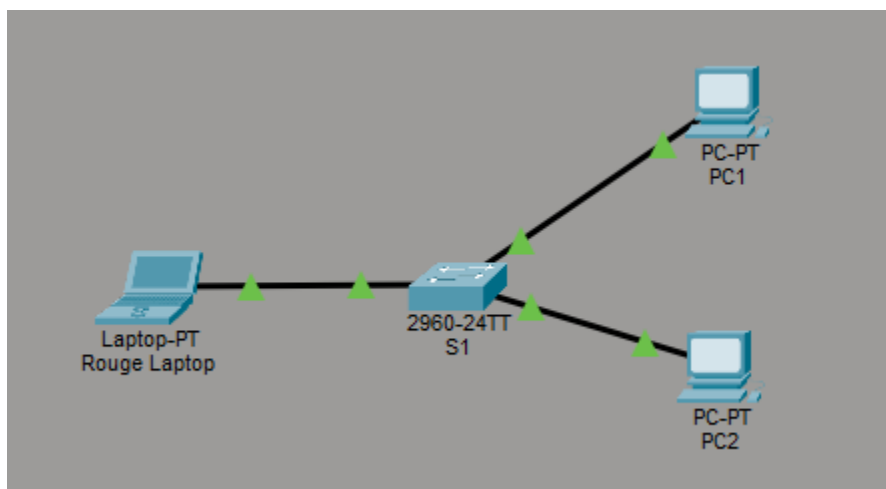


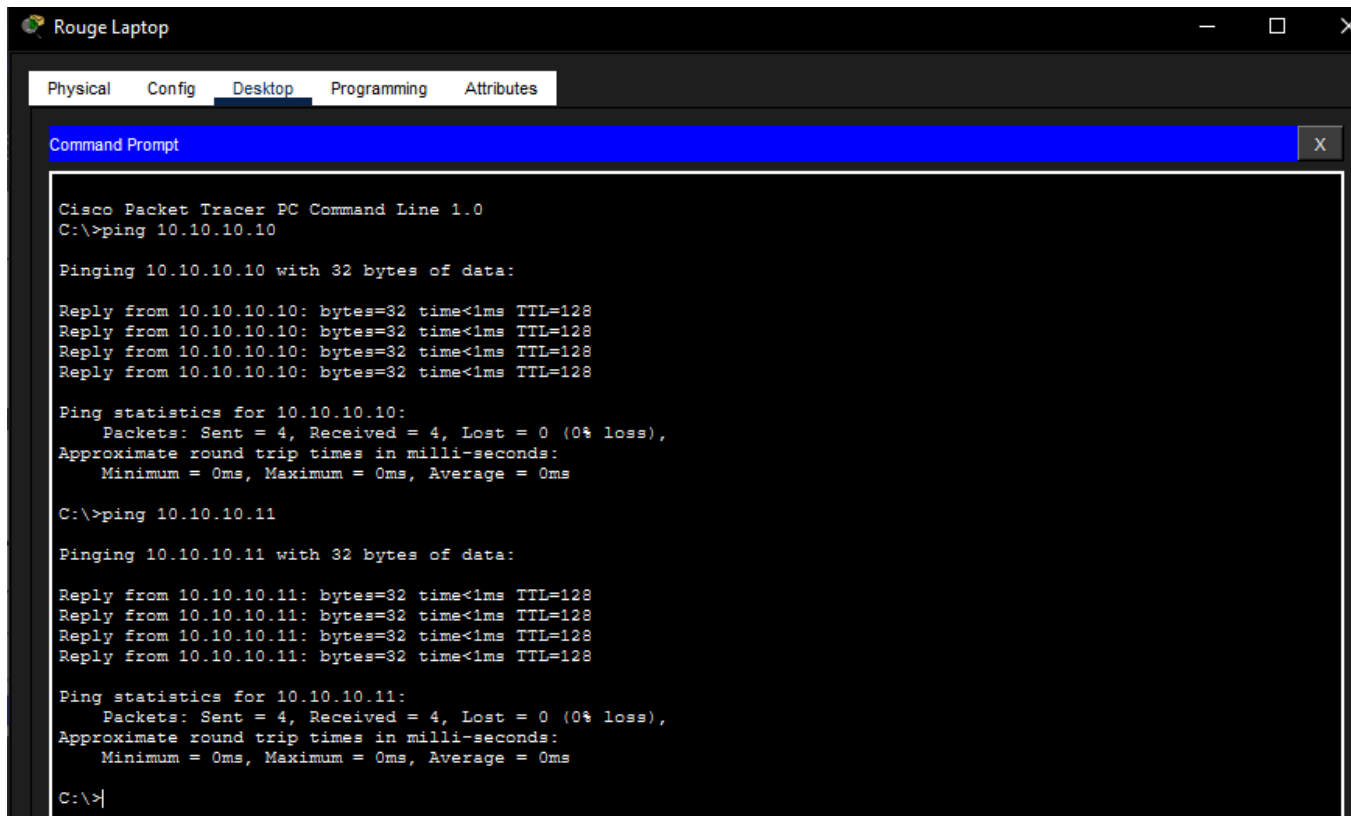
- f. Enable the port and verify that **Rogue Laptop** can ping **PC1** and **PC2**. After verification, shut down the port connected to **Rogue Laptop**.

```
S1(config)#int f0/3  
S1(config-if)#no shut
```

```
S1(config-if)#  
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
```





```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.11

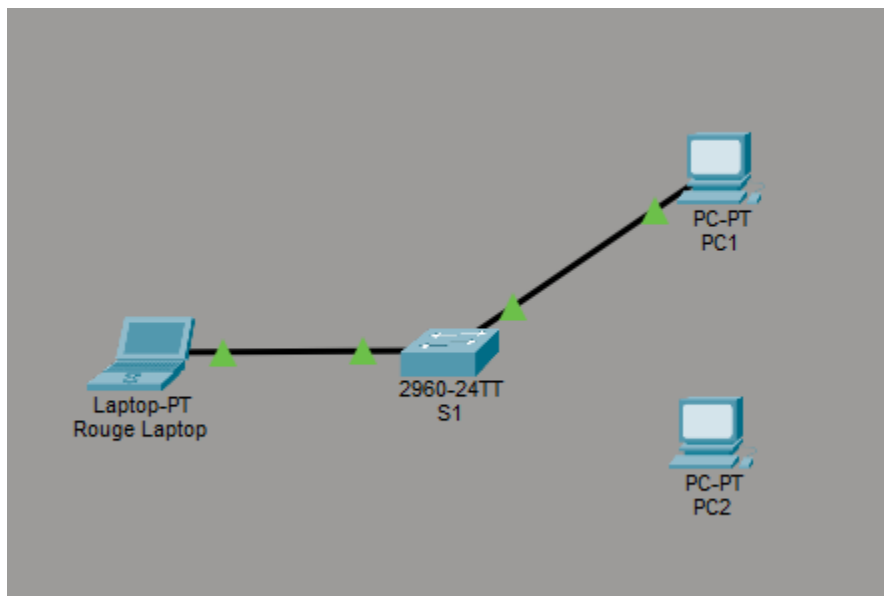
Pinging 10.10.10.11 with 32 bytes of data:

Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

- g. Disconnect **PC2** and connect **Rogue Laptop** to F0/2, which is the port to which PC2 was originally connected. Verify that **Rogue Laptop** is unable to ping **PC1**.



```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

- h. Display the port security violations for the port to which **Rogue Laptop** is connected.

```
S1# show port-security interface f0/2
```

How many violations have occurred?

4 Violation Count

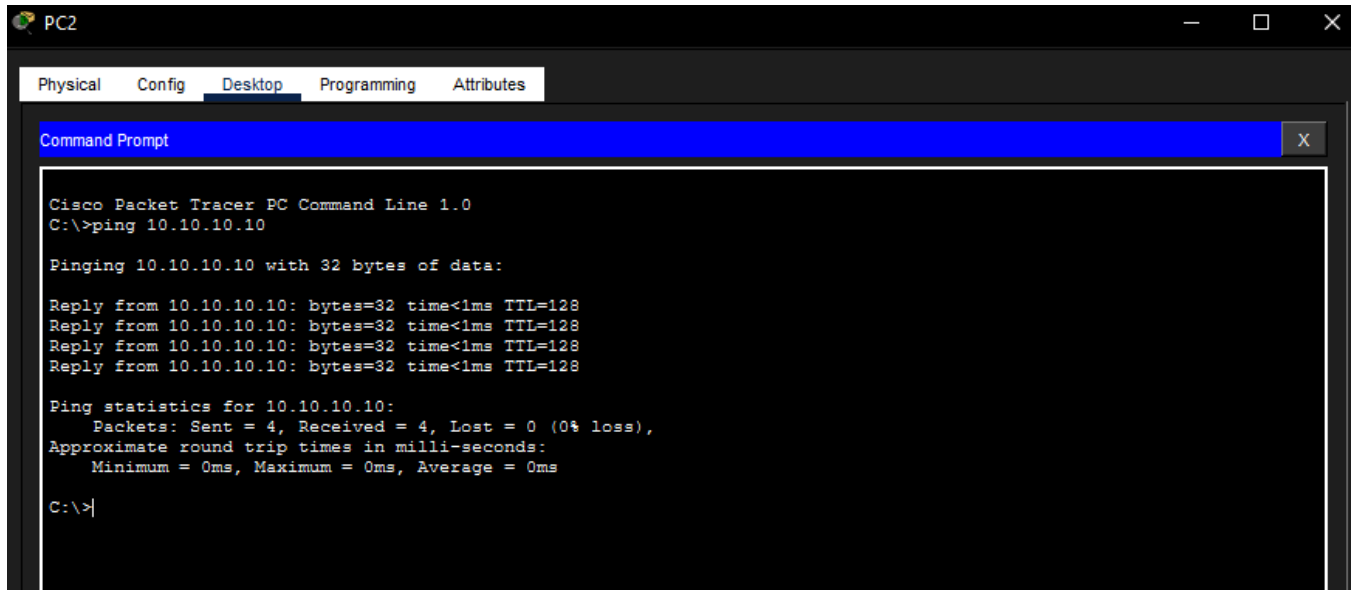
```
S1#show port-security int f0/2
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 000C.CF27.5EEC:1
Security Violation Count : 4
```

- i. Disconnect **Rogue Laptop** and reconnect **PC2**. Verify **PC2** can ping **PC1**.

Why is **PC2** able to ping **PC1**, but the **Rogue Laptop** is not?

PC2 can ping PC1 because Fa0/2 learned PC2's MAC address as the sticky secure MAC address. The Rogue Laptop cannot ping PC1 because it has a different MAC address. When Rogue Laptop is connected to Fa0/2, the switch detects a port-security violation and drops its traffic.

PC2 is able to communicate because interface F0/2 was configured with sticky port security restricted to a single MAC address. When PC2 connected initially, the switch dynamically learned and locked onto its MAC address. When the unauthorized Rogue Laptop was swapped in, it attempted to transmit data using a different MAC address. Since the interface violation mode was configured for "restrict," the switch kept the port up but dropped the unauthorized frames, generating log violations for each blocked ICMP request.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the Command Prompt is open. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```